

Using Large Language Models in Education Research

Annaliese Paulson

December 10, 2025

Roadmap

1. **Provide a framework for thinking about LLMs in social science and education research.**
2. Provide a conceptual overview of how LLMs work.
3. Discuss measurement validity in the context of LLMs.
4. Discuss methods to improve the quality of LLM output.

Types of Text and Images in Education

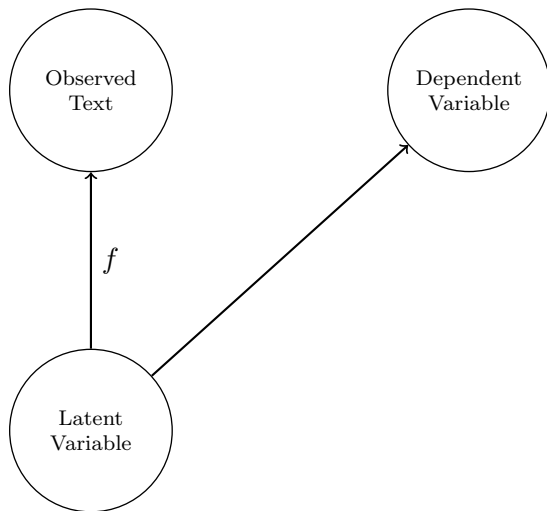
Thoughtfully used, LLMs have the potential to unlock new types of text and image data.

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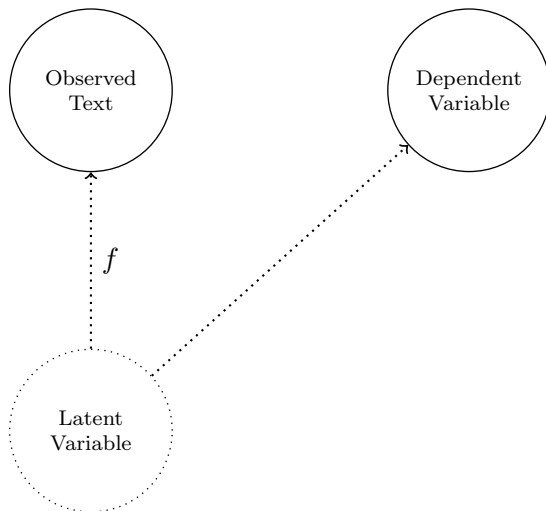
Thoughtfully used, LLMs have the potential to unlock new types of text and image data.

1. Measuring Admissions Essay Topics by Income (Alvero et al., 2021)
2. Measuring Race-Related Discourse in School Board Minutes (Anglin et al., 2025)
3. Measuring Skills of Majors in Job Board Postings (Hemelt et al., 2023)
4. Measuring Course Content in Syllabi (Biasi & Ma, 2022)
5. Measuring Teacher Instructional Practices from Transcripts of Classroom Instruction (Xu et al., 2024)
6. Measuring Local Economies of Colleges Using Satellite Images (Lehnert et al., 2024)
7. Measuring Race/Ethnicity of Children in Pictures from Children's Books (Adukia et al., 2023)

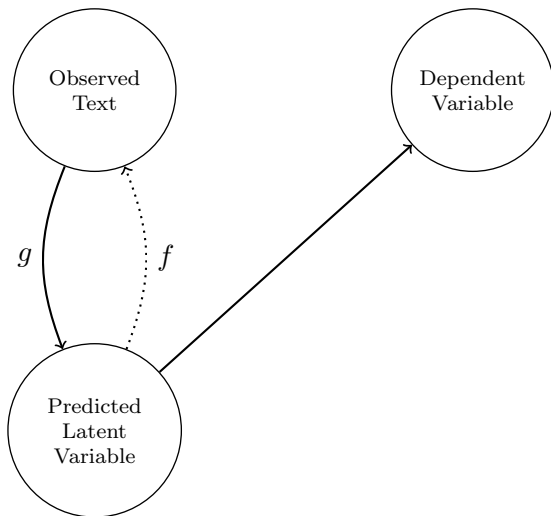
Thinking About Text and Images as Data



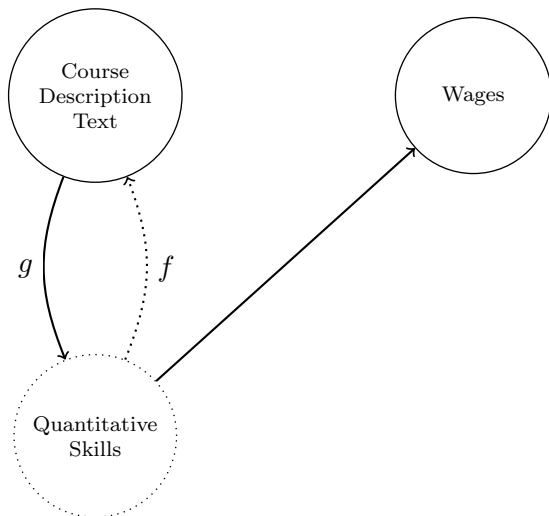
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Thinking About Text and Images as Data



Solving Missing Data Problems

Suppose you want to run the following regression using administrative data:

$$\log(Wages) = \beta CourseQuantSkills + \gamma \mathbf{X} + \epsilon$$

We need data like:

Student	Subject	Course	Quantitative Skills
10000	PSYC	201	True
10000	MATH	161	True
10000	ENGL	200	False
10000	HIST	221	False

Solving Missing Data Problems

But we typically have data like:

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We might use an LLM to fill in the missing data by predicting values with text from course descriptions.

Classification Prompts

Prompt:

Course Description:

{Insert Course Description}

Instruction:

Classify whether the university course description references teaching students quantitative skills like algebra, calculus, or statistics. Return only TRUE or FALSE.

Classification:

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Course Description:

PSYC 201: Statistics for Psychology - Statistics for Psychology is primarily a course that will introduce students to the application of statistics to the research process in psychology. Statistics are used to describe and to critically evaluate information. The two branches of statistics, descriptive and inferential statistics, will be covered in this course.

Instruction:

Classify whether the university course description references teaching students quantitative skills like algebra, calculus, or statistics. Return only TRUE or FALSE.

Classification:

LLM Response:

TRUE

Classification Prompts

Prompt:

Course Description:

MATH 160: Single Variable Calculus I - This course will provide an introduction to the topics and techniques of single-variable calculus. Differential calculus topics will include limits and derivatives of algebraic and transcendental functions as well as applications of the derivative. Integral calculus topics will include antiderivatives, area and the Fundamental Theorem of Calculus.

Instruction:

Classify whether the university course description references teaching students quantitative skills like algebra, calculus, or statistics. Return only TRUE or FALSE.

Classification:

LLM Response:

TRUE

Classification Prompts

Prompt:

Course Description:

ENGL 200: Personal and Public Writing - This course will explore ways writers move from personal, private writing into more public spheres. Narrative, ethnographic writing, representation, case studies, and genre are concepts which help students consider how writers situate themselves in their research in order to enter and intervene in ongoing public conversations. Students will bring rhetorical awareness to writing projects as they consider various strategies and audiences.

Instruction:

Classify whether the university course description references teaching students quantitative skills like algebra, calculus, or statistics. Return only TRUE or FALSE.

Classification:

LLM Response:

FALSE

Solving Missing Data Problems

Course	$g(Description)$ Pred. Quant Skills	Description
PSYC 201	True	Statistics for Psychology - Statistics for Psychology is primarily a course that will introduce students to the application of statistics to the research process in psychology. Statistics are used to describe and to critically evaluate information. The two branches of statistics, descriptive and inferential statistics, will be covered in this course.
MATH 161	True	Single Variable Calculus I - This course will provide an introduction to the topics and techniques of single-variable calculus. Differential calculus topics will include limits and derivatives of algebraic and transcendental functions as well as applications of the derivative. Integral calculus topics will include antiderivatives, area and the Fundamental Theorem of Calculus.
ENGL 200	False	Personal and Public Writing -This course will explore ways writers move from personal, private writing into more public spheres. Narrative, ethnographic writing, representation, case studies, and genre are concepts which help students consider how writers situate themselves in their research in order to enter and intervene in ongoing public conversations. Students will bring rhetorical awareness to writing projects as they consider various strategies and audiences.

Thinking About Text and Images as Data

- ▶ Text and images are inherently high-dimensional and non-tabular.

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- ▶ To use text and images as quantitative data, we need a function to map high-dimensional text to a small number of values. For ex:
 1. Map course description to binary measure of presence of quantitative skills
 2. Map school board meeting minutes to binary measures for reference to affirmative and oppositional discussions of race.

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 1. Map course description to binary measure of presence of quantitative skills
 2. Map school board meeting minutes to binary measures for reference to affirmative and oppositional discussions of race.
- ▶ Using text and images as quantitative data loses information and should only be used when human coding is infeasible due to scale.

When To Use Text and Images as Data

- ▶ Think of an LLM as a research assistant that is very diligent and never sleeps but also is not very skilled.
- ▶ LLMs are useful when we want to work with a large number of documents but don't need a rich understanding of them.

Roadmap

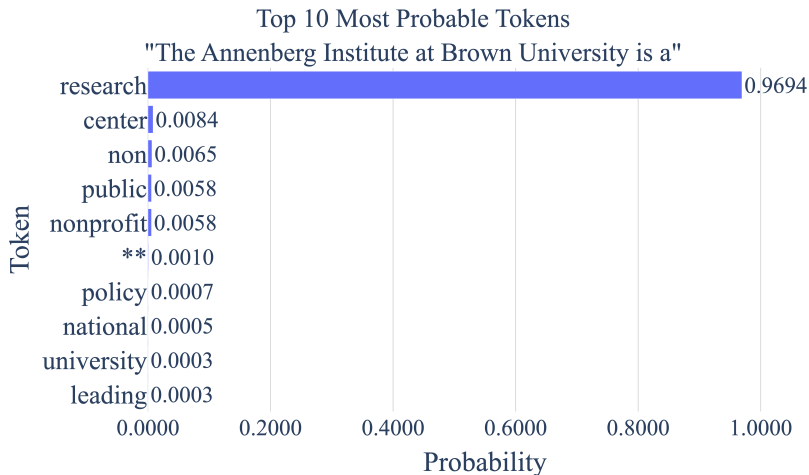
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2. **Provide a conceptual overview of how LLMs work.**
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What is a Language Model?

A language model is a probability distribution over possible next words, conditional on a sequence of preceding words.

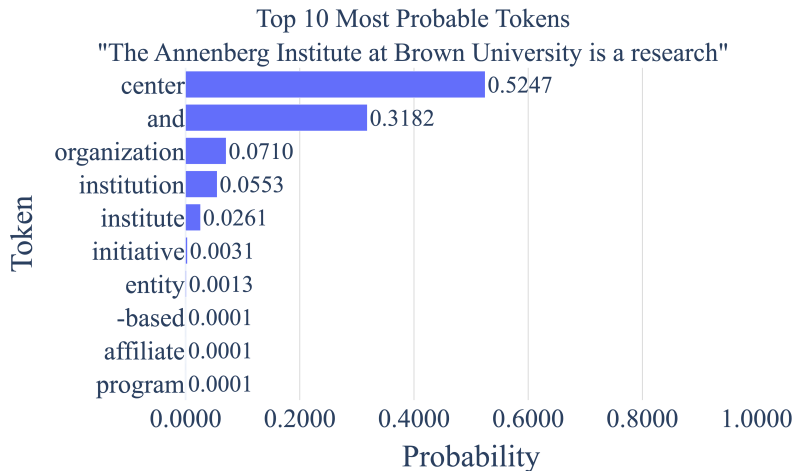
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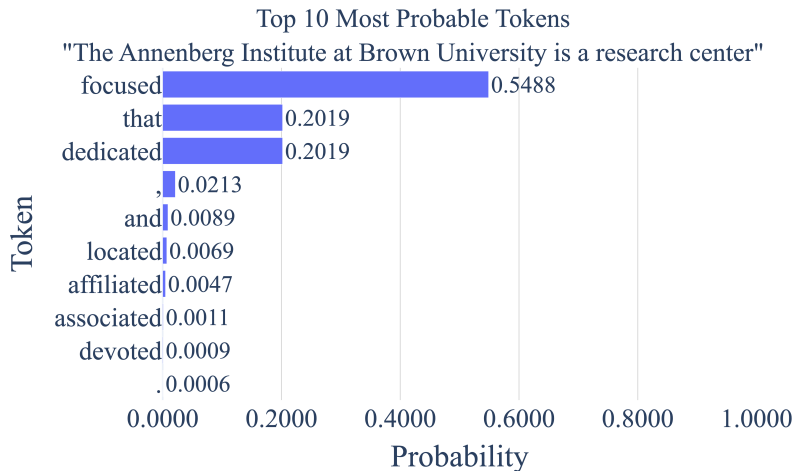
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What is a Language Model?

The Annenberg Institute is a research center focused on *education policy and practice*, particularly in the areas of *early childhood education, school leadership, and improving student outcomes*. It is part of the *Annenberg Foundation*, a philanthropy that supports initiatives aimed at enhancing the quality of education in the United States.

The Annenberg Institute is known for its work in:

- *Research and evaluation* of educational programs and policies.
- *Leadership development* for school administrators and educators.
- *Policy analysis* to inform decisions that support student success.
- *Public engagement* through reports, publications, and partnerships with schools and districts.

One of its most well-known initiatives is the *Annenberg Institute for School Reform*, which has worked with schools and districts across the country to improve teaching and learning, particularly in underserved communities.

If you're looking for more specific information about a particular program or initiative, feel free to ask! **<—im_end—>**

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$P(\text{"TRUE"} \mid \text{"Statistics for Psychology - Statistics for Psychology ..."}) \approx 1$

$P(\text{"FALSE"} \mid \text{"Statistics for Psychology - Statistics for Psychology ..."}) \approx 0$

Tokenization

LLMs represent text as tokens. Words (or subwords) and punctuation are assigned numeric numbers.

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The	An	nen	berg	Institute	at	Brown	University	is	a	research	center	.	<	im_end	>
976	1689	2542	9667	14544	540	18020	4923	382261	4176	6170	13	200266			

How Do LLMs Work?

1. User inputs a string.
 2. Model tokenizes string, converting text to sequence of numeric ids.
 3. Model represents each sequence of numeric ids as a dense numeric vector.
- ⋮ (Lots of Matrix Multiplication.)

N-2. Model outputs a probability distribution over tokens.
We add this token to the string and repeat the process until it outputs an end of sentence token.

- N-1. Model converts sequence of numeric token ids to string.
- N. Model outputs a string.

Lots of Matrix Multiplication

Local vs Proprietary Models

Local	Proprietary
Process data on a local machine	Send data to a for-profit company's servers to process
Generally smaller and run in local RAM	Generally larger and run on enterprise hardware
Typically up to 120 billion parameters	GPT-5 rumored to be trillions of parameters
Weights are publicly available	Weights are not available
No cost per token	Cost per token
Worse at knowledge tasks	Better at knowledge tasks

Training LLMs

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- ▶ Modern models are trained on trillions of tokens scraped from internet and books.
- ▶ This is very expensive (billions of \$) and requires lots of computing power!

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 - 3.2 Train model to predict Nth token using first through N-1st token.
 - ▶ See references for more details: Dell (2025), Jurafsky and Martin (2025)

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1. Clear evidence books are obtained illegally.
2. Model is trained on text from internet that seems to include every kind of problematic speech.
3. Clear evidence that models can be prompted to write problematic and illegal text.

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- ▶ Plenty of evidence LLMs encode the biases of society.
- ▶ It's not surprising that models trained on text generated by people learn the biases of society!
- ▶ Organizations try to mitigate bias from data after the fact but it is very challenging.

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Instruction Fine-Tuning

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- ▶ A probability distribution trained on things like Reddit isn't useful for many tasks.
- ▶ After training a model to generate text, we train it to follow instructions on a much smaller set of data like:

Instruction:

Given a tweet, classify it into one of 4 categories: Positive, Negative, Neutral, or Mixed.

Tweet:

The pumpkin was one of the worst that I've had in my life.

Classification:

Negative

Instruction Fine-Tuning

Prompt:

Course Description:

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LLM Response:

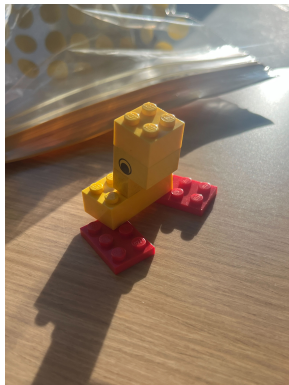
TRUE

Vision Models

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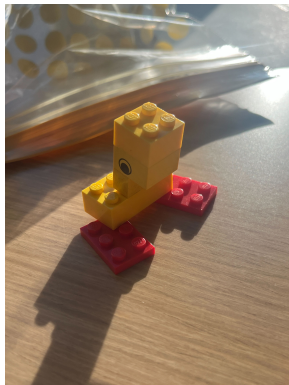


Prompt:

Describe the image.

Vision Models

Modern language models can be multimodal - they can use images to shape probability distribution.



Prompt:

Describe the image.

LLM Response:

This image shows a charming little LEGO creation of what appears to be a simple bird or duck character. It's made from yellow and red/pink LEGO bricks arranged in a minimalist design.

Vision Models

Extract the course descriptions from the course catalog page.

Return your analysis as a JSON array with elements containing these fields:

- course_description: [string containing the full course description, including subject code, catalog number, course content, and any administrative information about the course.']
- subject_code: [string containing the course's subject code. Like "MAT", "MATH", or "C S". Null if subject code is not present]
- catalog_number: [string containing the course's catalog number. Like "150", "1500A", or "1"]
- course_title: [string containing the course's title. Like "Calculus 1" or "American History Until 1865"]
- confidence: [low, medium, high]

Do not return additional text following the JSON.

Extract the course descriptions from this page and return a JSON:

Vision Models

PSYC 201 - Statistics for Psychology (3 credits)

Prerequisite: PSYC 100 and MATH 100 or higher (except First and Second Year Seminars and MATH 408); or consent of instructor
Statistics for Psychology is primarily a course that will introduce students to the application of statistics to the research process in psychology. Statistics are used to describe and to critically evaluate information. The two branches of statistics, descriptive and inferential statistics, will be covered in this course. Specific procedures that may be covered include measures of central tendency and variability, visual description of data, z-scores, correlation and linear regression, basic probability, parametric tests such as z-tests, t-tests, analysis of variance (ANOVAs), and non-parametric tests such as the chi-square test. (CQUR)

PSYC 215 - Service-Learning in Psychology (3 credits)

Prerequisite: Consent of instructor

Topics and partnerships with community agencies may vary from semester to semester. However, every semester students will have an opportunity to learn about some topic(s) in psychology, e.g., boys' development, girls' development, aging, mental illness, etc., and apply those concepts in work with a community partner. In regular class meetings students will read professional literature on the topic, reflect on that work in writing, discussions, media analyses and in other ways. In additional out-of-class experiences, students will engage in community service of some kind using what they have learned in class to inform the service work. Students will be active learners, and they will be encouraged to reflect on and evaluate the service work that they and their community partners do. The course may be repeated twice for a maximum of nine credits, though only the first three credits will count toward the psychology major.

```
{"subject_code": "PSYC",  
  "catalog_number": "201",  
  "course_title": "Statistics for Psychology",  
  "confidence": "high",  
  "course_description": "PSYC 201 - Statistics for Psychology (3  
    credits)  
  Prerequisite: PSYC 100 and ..."}  
}
```

Roadmap

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LLM Output Should Be Validated

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- ▶ A language model is a probability distribution over possible next words, conditional on a sequence of preceding words.
- ▶ LLMs are very good at producing coherent sounding text.
- ▶ But they sample probability distributions so can generate incorrect text known as hallucinations.

Hallucination

Prompt:

Write me a short essay on the use of technology and nudges in college student success. Highlight the important researchers in this space and include citations.

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Excerpt of LLM Response

Susan Dynarski and colleagues have extended this research, showing that personalized text message campaigns can improve course completion and GPA, particularly among students at risk of dropping out (Castleman & Meyer, 2020). Their work emphasizes the importance of timing, personalization, and actionable information in effective nudge design.

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Probabilistically, the model associates Sue with work on informational interventions.

The text misattributes work to her because it is sampling from a probability distribution.

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- ▶ Because models hallucinate, it's not clear how much we can trust output.
- ▶ As a general rule, we should treat the output of LLMs (and other machine learning models) with more suspicion than survey or administrative data.
- ▶ There's no reason to assume sampling from a probability distribution is true or useful in the social sciences!

Hallucination and Social Science

Prompt:

Course Description:

Introductory Psychology -

This is a survey of the different processes such as perception, sensation, learning and emotion, with a discussion of the underlying physiological processes as well as an introduction to the more complex areas such as personality development, psychopathology, social influences and testing. Methods of investigation and research will be integrated with the above topics.

Instruction:

Classify whether the university course description references teaching students quantitative skills like algebra, calculus, or statistics. Return only TRUE or FALSE.

Classification:

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Classification:

LLM Response:

TRUE

LLM Output Should Be Validated

- ▶ Text sampled from a probability distribution is not necessarily true.
- ▶ Before using a measure that uses text as data, we must present an argument for construct validity.
- ▶ Often, this means randomly sampling documents and annotating them yourself!

LLM Output Should Be Validated: Face Validity

- ▶ Face Validity - the measure seems plausible “on the face of it” to subject matter experts.
- ▶ Ex: faculty agree with LLMs classification of whether a course teaches quantitative skills.

LLM Output Should Be Validated: Convergent Validity

- ▶ Convergent Validity - our measure accords with other relevant measures.
- ▶ Ex: courses in economics classes teach more quantitative skills than anthropology.

LLM Output Should Be Validated: Hypothesis Validity

- ▶ Hypothesis Validity - the measure has expected relationships with other theoretical constructs.
- ▶ Ex: “chilly climate” in STEM departments hypothesizes we will see fewer women taking courses with quantitative skills.

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Improving Output

LLMs sample from a probability distribution, conditional on preceding text.

We can shift probability distribution by:

1. Restricting sampling: constrained generation.
2. Providing examples: one- and few-shot learning.
3. Instructing model to “think step by step”: chain of thought.
4. Fine-tuning model on our own data.

Constrained Generation

Often, we want a model to produce specific output:

1. $CourseQuantSkills \in \{True, False\}$
2. $StudentSentiment \in \{Positive, Negative, Mixed, Neutral\}$

A simple solution called constrained generation: change the final probability distribution for all words not in a set to zero.

One and Few-Shot Learning

We can shift the probability distribution by providing examples.

One and Few-Shot Learning

We can shift the probability distribution by providing examples. Prepending an example to our prompt, shifts the Intro to Psychology classification to “FALSE”

Course Description:

Qualitative Methods and Urban Ethnography - This course introduces students to qualitative research methods in social research. The course will consider the historical and theoretical background to qualitative methods, the principles of qualitative methods, as well as particular methods including participant observation, qualitative interviews, and the use of documents.

Instruction:

Classify whether the university course description references teaching students quantitative skills like algebra, calculus, or statistics. Return only TRUE or FALSE.

Classification:

FALSE

Chain-of-Thought

Prompting models to “think step by step” improves performance.

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Instruction:

Classify whether the university course description references teaching students quantitative skills like algebra, calculus, or statistics. Let's think step by step. Ultimately, return CLASSIFICATION: followed by TRUE or FALSE.

Chain-of-Thought

Prompting models to “think step by step” improves performance.

Instruction:

Classify whether the university course description references teaching students quantitative skills like algebra, calculus, or statistics. Let's think step by step. Ultimately, return CLASSIFICATION: followed by TRUE or FALSE.

Excerpt of LLM Response

Step 1: Read the course description carefully.

Step 2: Identify whether quantitative skills (algebra, calculus, or statistics) are mentioned.

The course description does not mention algebra, calculus, or statistics explicitly. While it refers to "methods of investigation and research," this is a general statement and does not specifically indicate the teaching of quantitative skills like statistics.

CLASSIFICATION: FALSE

Fine-Tuning

- ▶ It's not clear that increasing scale is necessary for social science research (Dell, 2025; Ziems et al., 2024).
- ▶ LLMs like GPT-5 try to solve every problem with one probability distribution.
- ▶ As researchers, we don't need our models to write recipes or poems. We need them to excel at specific tasks.

Instruction Fine-Tuning for Tasks

Rather than rely on billion or trillion parameter models, we can:

1. Curate our own data by annotating a random sample of text.

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2. Download a small open-source model.
3. Train our small model for a domain specific task.
4. Evaluate our model on new data.

Benefits of Fine-tuning Smaller Models

Small models:

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4. can surpass large models performance with domain specific training.

Fine-tuning and Supervised Learning

Rather than prompt a generative model, we take the weights learned during pretraining and use them in supervised classification.

Classification models have a number of benefits:

1. Smaller less computationally expensive, and cheaper than generative LLMs.
2. Locally hosted.
3. Reproducible over time.
4. Generally perform better on specific tasks! (Dell, 2025; Ziems et al., 2024)

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2. Language models are probability distributions of next tokens. They can be useful but they do not need to be true.
3. All LLM output should be validated before being used in research.
4. For-profit language models are very large and inefficient for researchers. By leveraging domain expertise and fine-tuning we can do higher quality research with less computing resources.

Learning More

LLMs in Social Science Research:

Melissa Dell is a leader in applying LLMs in social science. In addition to this paper, she posts her lectures and course materials online at this knowledge base.

- ▶ Dell, M. (2025). Deep Learning for Economists. *Journal of Economic Literature*, 63(1), 5–58. <https://doi.org/10.1257/jel.20241733>
- ▶ Ziems, C., Held, W., Shaikh, O., Chen, J., Zhang, Z., & Yang, D. (2024). Can Large Language Models Transform Computational Social Science? *Computational Linguistics*, 50(1), 237–291. https://doi.org/10.1162/coli_a_00502

Natural Language Processing:

- ▶ Jurafsky, D., & Martin, J. H. (2025). *Speech and language processing: An introduction to natural language processing, computational linguistics, and speech recognition, with language models* (3rd ed.). <https://web.stanford.edu/~jurafsky/slp3/>

Using LLMs in Python:

Hugging Face is the open-source company that provides the packages for LLM usage in python. They provide a number of wonderful resources including this course to begin working with LLMs.

- ▶ Hugging Face. “The Hugging Face Course, 2022.” 2022. <https://huggingface.co/course>.

Using Text as Data:

- ▶ Grimmer, J., Roberts, M. E., & Stewart, B. M. (2022). *Text as data: A new framework for machine learning and the social sciences*. Princeton University Press.

Examples Using Text and Images as Data

- ▶ Adukia, A., Eble, A., Harrison, E., Runesha, H. B., & Szasz, T. (2023). What we teach about race and gender: Representation in images and text of children's books. *The Quarterly Journal of Economics*, 138(4), 2225–2285. <https://doi.org/10.1093/qje/qjad028>
- ▶ Alvero, A., Giebel, S., Gebre-Medhin, B., Antonio, A. I., Stevens, M. L., & Domingue, B. W. (2021). Essay content and style are strongly related to household income and SAT scores: Evidence from 60,000 undergraduate applications. *Science Advances*, 7(42), eabi9031. <https://doi.org/10.1126/sciadv.abi9031>
- ▶ Anglin, K. L., Christian, A., Jacob, B. A., & Singleton, J. D. (2025). *Education governance and race: An analysis of school board discourse using large language models* (EdWorkingPaper No. 25-1175). Annenberg Institute at Brown University. <https://doi.org/10.26300/EW42-T954>
- ▶ Biasi, B., & Ma, S. (2022). *The education-innovation gap* (Working Paper No. 29853). National Bureau of Economic Research. <http://www.nber.org/papers/w29853>
- ▶ Hemelt, S. W., Hershbein, B., Martin, S., & Stange, K. M. (2023). College majors and skills: Evidence from the universe of online job ads. *Labour Economics*, 85, 102429. <https://doi.org/10.1016/j.labeco.2023.102429>
- ▶ Lehnert, P., Dell, M., Backes-Gellner, U., & Bettinger, E. (2024). *The effect of postsecondary educational institutions on local economies: A bird's-eye view* (Working Paper No. 32679). National Bureau of Economic Research. <https://www.nber.org/papers/w32679>
- ▶ Xu, P., Liu, J., Jones, N., Cohen, J., & Ai, W. (2024). The promises and pitfalls of using language models to measure instruction quality in education. In *Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)* (pp. 4375–4389). Association for Computational Linguistics. <https://doi.org/10.18653/v1/2024.naacl-long.246>